Code No: 132AC **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech I Year II Semester Examinations, September - 2023
MATHEMATICS - III
(Common to CE, EEE, ME, ECE, CSE, EIE, IT, MCT, MMT, AE, MIE)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a) The probability density function is $y = \begin{cases} K(3x^2-1), -1 \le x \le 2 \\ 0, \quad elsewhere \end{cases}$. Find K. [2]

b) If for a poisson variate $2p(x=0) = p(x=2)$. Find mean. [3]

c) Write the formula to calculate F to test the difference the between variances. [2]

d) In a random sample of 200 packages shipped by air freight 52 had some damage. Find the maximum error with 95% confidence of proportions. [3]

e) Write the formula to find the maximum error for:
i) single proportions ii) difference between the proportions [2]

f) In a random sample of 100 packages shipped by air freight 13 had some damage. Find the standard error of proportions. [3]

g) If first approximation of a root of $x^2 - x - 4 = 0$ is $x_0 = 2$ then find x_1 by Newton Raphson method. [2]

h) Find the two points a and b such that $x\sin x + \cos x = 0$ has a root in between a and b. [3]

i) Write a formula to find the approximate value of
$\int_a^b f(x)dx$, when $(x_i, f(x_i))\, i = 0,1,2,3,....n$ are known and $a = x_0 < x_1 < x_2 < ..... < x_n = b.$ [2]

j) If $y' = 2x - y$ y(1)=3 find the solution, up to third degree term, using Picard's method. [3]

PART – B

(50 Marks)

2.a) Find the mean and the variance of uniform probability distribution given by
f(x) = $\frac{1}{n}$ for x= 1, 2, 3, 4…….. n

b) Ten coins are thrown simultaneously. Find the probability of getting: [5+5]
i) At least 1 head ii) At least seven heads

OR

3.a) Assume that 50% of the Engineers are good in Mathematics. Find the probability that among 9 engineers.
i) Exactly 5 ii) At least 2 are good in Mathematics.

b) Given that the mean heights of students in a class is 158 cms with a standard deviation of 20 cms. Find how many students heights lie
i) Between 150 cms and 170 cms.
ii) Above 180 cms. [5+5]

4. Five unbiased dice were thrown 276 times and the number of times 4 or 5 or 6 was obtained is given below.

Number appeared on the die	1	2	3	4	5	6
Frequency	40	32	29	59	57	59

Fit a suitable distribution and test whether the die is unbiased. [10]

OR

5. The gain in weight of two random samples of rats fed on two different diets A and B are given below. Examine whether the difference in mean weight increases is significant or not. [10]

Diet A	13	14	10	11	12	16	10	8	-
Diet B	7	10	12	8	10	11	9	10	11

6.a) A large group of athletes is found to have a mean weight of 140 lbs with a standard deviation of 5 lbs. One student from a college was found to weight 120 lbs. Can it be reasonably concluded that he was not an athlete?

b) Among the items produced by a factory out of 500, 15 were defective, in another sample out of 400, 20 were defective. Test the significance between the difference of two proportions at 5% level. [5+5]

OR

7.a) The mean height of 40 male students is 68 inches with a standard deviation of 3 where as the mean height of 50 female students is 66 inches with a standard deviation of 2.5 inches. Test the significance between the difference of two means at 5% level.

b) Random samplers of 300 men and 500 women are asked whether they would like to have a fly over near their residence 150 and 200 women were in favour of the proposal. Find 95% confidence interval for the difference of proportions. [5+5]

8.a) Find a real root of $\cos x - x^2 - x = 0$ using Newton Raphson method.

b) Fit a parabola of the form $y = a + bx + cx^2$ for the following data. [5+5]

x	10	12	15	23	20
y	14	17	23	25	21

OR

9. Solve the system of equations using Gauss Seidal method.

$$8x - 3y + 2z = 20$$
$$6x + 3y + 12z = 35$$
$$4x + 11y - z = 33$$

[10]

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10.a) Find the first two derivatives of f(x) at x= 1.2 from the following table.

x	1.0	1.1	1.2	1.3	1.4	1.5	1.6
y	7.989	8.403	8.781	9.129	9.451	9.75	10.031

b) Evaluate $\int_0^1 \frac{dx}{1+x^2}$ by taking $h = \frac{1}{6}$ using Simpson's $\frac{1}{3}rd$ rule. [5+5]

OR

11. Find y(0.1) and y(0.2) using Runge Kutta method given that $y' = x^2 - y, y(0) = 1$. [10]

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